

PROJECT APEX

Enterprise LNG Operations Intelligence Platform

VOLUME 5

Portfolio, Documentation & Interview Preparation

LinkedIn Write-Up, GitHub README, Interview Q&A, and Presentation Script

PART OF THE SERIES

Atlantic LNG Operations Limited (ALOL) — Power BI Project

Prepared by

Allwell Hart

Data Analyst / Power BI Developer

Series Overview

Volume 1 — Project Planning & Business Understanding

Volume 2 — Data Preparation & Data Modeling

Volume 3 — DAX Development

Volume 4 — Dashboard Design & Storytelling

Volume 5 — Portfolio, Documentation & Interview Preparation (this document)

Prepared June 2026

Table of Contents

Table of Contents	2
1. Introduction	4
2. Complete Project Documentation Summary	5
2.1 Project Identity.....	5
2.2 Dataset Summary	5
2.3 DAX Measure Library Summary	5
2.4 Dashboard Summary.....	6
3. Business Insights to Discover	7
3.1 Production Insights	7
3.2 Export & Commercial Insights.....	7
3.3 Maintenance & Reliability Insights	7
3.4 HSE Insights.....	8
4. LinkedIn Portfolio Description	9
4.1 LinkedIn Project Entry	9
4.2 LinkedIn Post Announcement (Optional).....	9
5. GitHub README Template.....	10
6. Interview Preparation Guide.....	11
6.1 Data Modeling Questions	11
6.2 DAX Questions	13
6.3 Dashboard Design Questions	14
6.4 Business Understanding Questions	15
7. Full Presentation Script.....	16
7.1 Opening (1–2 minutes)	16
7.2 Executive Dashboard (2–3 minutes)	16
7.3 Production Dashboard (2–3 minutes).....	16
7.4 Shutdown & Finance (3–4 minutes).....	17
7.5 Closing (1–2 minutes)	17
8. Project APEX Completion Checklist.....	18
8.1 Volume 1 — Project Planning	18
8.2 Volume 2 — Data Modeling.....	18
8.3 Volume 3 — DAX Development	18
8.4 Volume 4 — Dashboard Design	19
8.5 Volume 5 — Portfolio & Documentation.....	19

1. Introduction

Volume 5 is the final volume of the Project APEX series. The analytical work is complete — the data is modeled, the DAX is written, and the dashboards are designed. What remains is packaging all of that work into something that can be seen, shared, and discussed professionally: a portfolio-ready project that proves both technical competence and business understanding to any employer, recruiter, or client who reviews it.

This volume delivers five things: a complete project documentation summary, a LinkedIn portfolio write-up ready to copy and adapt, a GitHub README template for the project repository, a full interview preparation guide with likely questions and model answers, and a presentation script for live demonstrations of the dashboard.

2. Complete Project Documentation Summary

This section provides a one-stop summary of the entire Project APEX engagement — what was built, what decisions were made, and what was learned. It is written in a format suitable for inclusion in a professional portfolio document or a project handover pack.

2.1 Project Identity

Attribute	Detail
Project Name	Project APEX — Enterprise LNG Operations Intelligence Platform
Company Simulated	Atlantic LNG Operations Limited (ALOL) — Fictional
Developer	Allwell Hart, Data Analyst / Power BI Developer
Project Type	End-to-end Power BI portfolio project — self-directed
Domain	Oil & Gas / LNG Operations
Period Covered	2019–2024 (6-year operational simulation)
Tools Used	Python (data generation), Power BI Desktop, Power Query, DAX, Microsoft Word
Delivery Format	5-volume documentation series + Power BI .pbix file + CSV dataset
Volume Count	5 volumes, approximately 120 pages of professional documentation in total

2.2 Dataset Summary

Category	Detail
Total tables	14 (7 dimension, 7 fact)
Total records	~143,000 rows across all fact tables
Dimension tables	dim_calendar, dim_trains, dim_products, dim_countries, dim_customers, dim_equipment, dim_employees
Fact tables	fact_production, fact_export_shipments, fact_maintenance, fact_shutdown_events, fact_hse_incidents, fact_finance, fact_weather
Referential integrity	19 relationships, all validated — zero orphaned keys
Schema type	Star schema — optimized for Power BI VertiPaq engine
Date range	2019-01-01 to 2024-12-31
Special modeling note	fact_shutdown_events uses dual-date (StartDateKey active, EndDateKey inactive) with USERELATIONSHIP() in DAX

2.3 DAX Measure Library Summary

Measure Table	Measure Count	Key Measures
_Production Measures	12	Total LNG Produced, Availability %, Utilization %, Conversion Efficiency %, LNG YoY %
_Export Measures	11	Total Export Revenue, Customer Concentration %, Top Customer, Revenue YoY %
_Maintenance Measures	8	Corrective Maintenance Share %, Work Order Completion Rate %, Equipment Reliability Score
_Shutdown Measures	8	Unplanned Shutdown Share %, Revenue Lost to Shutdowns, USERELATIONSHIP() end-date measure
_HSE Measures	5	Total Incidents, LTIFR, HSE Incident YoY % (reversed convention)
_Finance Measures	9	Profit Margin %, OPEX as % of Revenue, Revenue per Tonne LNG, Profit YTD
_Weather Measures	4	High Wind Day Count, Total Rainfall, Average Temperature

2.4 Dashboard Summary

Page	Title	Primary Visual Type	Signature Feature
1	Executive Dashboard	KPI cards + dual-axis line	Single-screen business health summary
2	Production Dashboard	Train heatmap matrix + line chart	Train-by-train performance comparison
3	Export Dashboard	Revenue bar + customer table	Customer concentration risk card
4	Finance Dashboard	Waterfall + profitability trend	Revenue-to-profit bridge chart
5	Maintenance Dashboard	Scatter + matrix heatmap	Corrective vs preventive cost split
6	Shutdown Dashboard	Gantt timeline + KPI cards	USERELATIONSHIP() end-date KPI card
7	HSE Dashboard	Stacked severity bar + area chart	Reversed YoY conditional formatting
8	Weather Dashboard	Multi-line + scatter correlation	Rainfall vs availability correlation test
9	Map Dashboard	Bubble world map	Lat/Lon-based trade flow visualization

3. Business Insights to Discover

This section documents the specific analytical discoveries that the Project APEX dashboard suite is designed to surface. In a real engagement, these would be the ‘findings’ section of a final report to the client. In a portfolio context, being able to articulate what your dashboard actually reveals (not just that it exists) is one of the clearest demonstrations of analytical maturity.

3.1 Production Insights

- Train availability is the primary driver of LNG output variance year-over-year — trains with average availability above 93% consistently outperform capacity targets while those dropping below 85% lose disproportionately more production than the availability percentage alone suggests, due to restart inefficiencies
- The heatmap matrix (Production Dashboard) visually confirms that planned turnaround months show sharp production dips in specific cells, with a visible recovery ramp-up in the following month as trains return to full output after restart
- Utilization consistently tracks slightly below Availability in the simulation, reflecting the realistic constraint that a train being ‘available’ does not mean it is immediately running at full nameplate capacity

3.2 Export & Commercial Insights

- Customer concentration risk is visible: the top 5 customers by revenue account for a disproportionate share of total export revenue, which an LNG company in real life would actively manage through contract diversification
- Europe and Asia consistently dominate the regional revenue split, reflecting the real-world LNG trade pattern where Atlantic basin LNG flows primarily to European spot markets (especially post-2022) and contracted Asian off-takers (Japan, Korea, China)
- Spot cargo pricing (ShipmentStatus-based analysis of Spot vs Long-term SPA customers) reveals the premium or discount applied to unsold spot cargoes versus contracted volumes — a direct commercial performance indicator

3.3 Maintenance & Reliability Insights

- Corrective maintenance spend as a share of total maintenance cost tends to rise in the year following a period of below-average production availability — suggesting that trains which underperform in one period accumulate a deferred maintenance backlog that surfaces as reactive spend in the next period
- Critical A and Critical B equipment items account for the majority of both maintenance cost and downtime hours despite being a minority of total equipment count — consistent with the real-world Pareto principle in industrial maintenance

- The Equipment Reliability Score scatter chart identifies specific equipment items that are simultaneously high-cost to maintain and frequently involved in HSE incidents, which are the priority candidates for capital replacement decisions

3.4 HSE Insights

- Near Miss incidents significantly outnumber all other severity categories, which in real HSE management is considered a positive leading indicator: a workforce that consistently reports near misses has a stronger safety culture than one where near misses go unrecorded but serious incidents occur
- Process Area A and Process Area B consistently rank as the highest-incident areas in the simulation — consistent with these being the closest areas to active liquefaction operations where the highest process risk concentration exists
- The HSE incident YoY trend (in the simulation) shows year-on-year variation that can be correlated against maintenance spend changes — periods of reduced maintenance investment correlate weakly with increased incident frequency, supporting the real-world business case for preventive maintenance

4. LinkedIn Portfolio Description

The text below is a complete LinkedIn project description, ready to post in the Featured section or Projects section of your LinkedIn profile. It is written to be read in under 60 seconds by a recruiter or hiring manager who may spend less than one minute on each portfolio item.

4.1 LinkedIn Project Entry

Project APEX — Enterprise LNG Operations Intelligence Platform

Post this as the project title, then use the following as the description body:

An end-to-end Power BI analytics project simulating the full operations of Atlantic LNG Operations Limited (ALOL), a fictional LNG producer modeled on Bonny Island, Nigeria. **What I built:** • Engineered a 14-table star schema (~143,000 records) covering 6 years of LNG production, exports, maintenance, HSE incidents, shutdowns, finance, and weather data • Performed full Power Query data cleaning and model validation with zero orphaned foreign keys across 19 relationships • Developed 45+ DAX measures across 7 subject-area measure tables, including time intelligence, USERELATIONSHIP() for dual-date patterns, and cross-fact-table analytical measures • Designed 9 dashboard pages (Executive, Production, Export, Finance, Maintenance, Shutdown, HSE, Weather, Map) with consistent UI/UX, custom navigation, drill-through, and tooltip design • Documented the entire project in a 5-volume professional series covering business context, data modeling, DAX development, and dashboard design **Key skills demonstrated:** Power BI • Power Query (M) • DAX • Star Schema Design • Oil & Gas Domain Knowledge • Data Storytelling • KPI Framework Design • Dashboard UX

4.2 LinkedIn Post Announcement (Optional)

If you want to announce this project as a standalone LinkedIn post when you publish it, use this template:

Just completed Project APEX — a full end-to-end Power BI project simulating enterprise LNG operations for a fictional oil & gas company. 6 years of data. 14 tables. 45+ DAX measures. 9 dashboards. 5 volumes of professional documentation. From data engineering to star schema modeling, from USERELATIONSHIP() patterns to a full UI/UX design system, this project covers the entire BI lifecycle from raw data to executive-ready analytics. If you work in energy, data analytics, or Power BI — I'd love to hear your thoughts. Link to the full project in the comments. #PowerBI #DAX #DataAnalytics #LNG #OilAndGas #DataVisualization #BusinessIntelligence

5. GitHub README Template

If you publish the Project APEX CSV files, Power BI file, and documentation on GitHub, the README below is a complete, professional project description. Copy it into a file named README.md in your repository root.

```
# Project APEX – Enterprise LNG Operations Intelligence Platform > An end-to-end Power BI
portfolio project simulating the full > operational analytics of Atlantic LNG Operations
Limited (ALOL), > a fictional LNG company modeled on Bonny Island, Nigeria. **Author:**
Allwell Hart – Data Analyst / Power BI Developer **Domain:** Oil & Gas / LNG Operations
**Period Simulated:** 2019 – 2024 (6 years) **Status:** Complete --- ## What Is This
Project? Project APEX is a self-directed, portfolio-grade Power BI project that simulates
the enterprise analytics platform of a fictional LNG operator. It demonstrates the full BI
lifecycle: - Dataset engineering (Python-generated star schema, ~143,000 rows) - Power
Query data cleaning and model validation - DAX measure development (45+ measures, 7
measure tables) - Dashboard design (9 pages, professional UI/UX system) - Business
storytelling and documentation (5-volume series) --- ## Repository Structure APEX/
data/ <- 14 CSV files (star schema) dim_calendar.csv
dim_trains.csv dim_products.csv dim_countries.csv dim_customers.csv
dim_equipment.csv dim_employees.csv fact_production.csv
fact_export_shipments.csv fact_maintenance.csv fact_shutdown_events.csv
fact_hse_incidents.csv fact_finance.csv fact_weather.csv docs/
<- 5-volume documentation series (.docx) ALOL_Volume1_Project_Planning.docx
ALOL_Volume2_Data_Prep_Modeling.docx ALOL_Volume3_DAX_Development.docx
ALOL_Volume4_Dashboard_Design.docx ALOL_Volume5_Portfolio_Interview.docx pbix/
APEX_Dashboard.pbix <- Power BI file README.md --- ## Data Model 14-table star
schema | 19 relationships | 0 orphaned keys Dimension tables: dim_calendar, dim_trains,
dim_products, dim_countries, dim_customers, dim_equipment, dim_employees Fact tables:
fact_production, fact_export_shipments, fact_maintenance, fact_shutdown_events,
fact_hse_incidents, fact_finance, fact_weather Special pattern: fact_shutdown_events
uses dual DateKey columns (StartDateKey [active] + EndDateKey [inactive]) with
USERRELATIONSHIP() in DAX for end-date filtering. --- ## How to Use 1. Clone or download
this repository 2. Open APEX_Dashboard.pbix in Power BI Desktop 3. If prompted to locate
data sources, point Power BI to the /data/ folder in this repository 4. Refresh the
data model 5. All 9 dashboard pages will be available in the report --- ## Dashboard
Pages | Page | Title | Key Feature | |-----|-----|-----| | 1 | Executive
Dashboard | Plant-wide KPI summary | | 2 | Production Dashboard | Train heatmap matrix | |
3 | Export Dashboard | Customer concentration risk | | 4 | Finance Dashboard | Revenue-to-
profit waterfall | | 5 | Maintenance Dashboard | Corrective vs preventive split | | 6 |
Shutdown Dashboard | Gantt timeline + USERRELATIONSHIP | | 7 | HSE Dashboard | Reversed YoY
conditional format | | 8 | Weather Dashboard | Marine wind threshold analysis | | 9 | Map
Dashboard | Global trade flow bubble map | --- ## Documentation The /docs/ folder
contains a 5-volume professional documentation series covering the entire project
lifecycle. Each volume is a standalone Word document: Volume 1 – Project Planning &
Business Understanding (~27 pages) Volume 2 – Data Preparation & Data Modeling (~20 pages)
Volume 3 – DAX Development (~26 pages) Volume 4 – Dashboard Design & Storytelling (~30
pages) Volume 5 – Portfolio, Documentation & Interview Prep (~28 pages) --- ## Contact
Allwell Hart Data Analyst / Power BI Developer Location: Port Harcourt, Rivers State,
Nigeria
```

6. Interview Preparation Guide

This section provides model answers to the interview questions most likely to arise when discussing Project APEX with a hiring manager, technical lead, or recruiter. Each answer is written to be approximately 60–90 seconds long when spoken aloud — long enough to demonstrate genuine depth, short enough not to lose the interviewer. Every answer connects directly to a specific decision made in Volumes 1–4.

6.1 Data Modeling Questions

Q: Walk me through the data model you built for this project.

A: The model is a 14-table star schema with 7 dimension tables and 7 fact tables, covering production, exports, maintenance, shutdowns, safety incidents, finance, and weather for a simulated LNG company over six years. All 7 fact tables connect to a shared date dimension — `dim_calendar` — which is the backbone of all time intelligence in the model. I built and validated 19 relationships with zero orphaned foreign keys across all tables. The one non-standard element is `fact_shutdown_events`, which has two date keys — a start date and an end date — because a shutdown event spans a period rather than occurring on a single day. I handled that by making the start date the active relationship and the end date inactive, using `USERELATIONSHIP()` in DAX when I specifically needed to filter by when a shutdown ended.

Q: Why did you choose a star schema over a normalized or snowflake design?

A: Three reasons. First, Power BI's VertiPaq engine is specifically optimized for star schemas — column-store compression and DAX filter context propagation both perform significantly better against a flat star than against multiple joined tables. Second, a star schema with single-direction filter flow is much easier to reason about when writing DAX, because you always know where filters are coming from. And third, it matches Microsoft's own documented best practice for Power BI semantic models. The one place where my model has a dimension referencing another dimension — `dim_customers` referencing `dim_countries` via `CountryID` — I validated carefully that filters still propagate correctly and documented the reason for that design choice in the project documentation.

Q: How did you handle the dual-date relationship on the shutdown events table?

A: In Power BI you can only have one active relationship between any two tables. `fact_shutdown_events` has both a `StartDateKey` and `EndDateKey`, both referencing `dim_calendar`. I made `StartDateKey` the active relationship because most shutdown analytics are start-date-based — you want to know when a shutdown began. I then created a second, inactive relationship on `EndDateKey`. In DAX, whenever I need to filter shutdown events by when they ended — for example, counting shutdowns that concluded within the current quarter — I use `USERELATIONSHIP()` inside a `CALCULATE()` function to explicitly activate the inactive relationship for that specific measure only. This is a very common real-world modeling pattern and one of the things interviewers tend to test on specifically.

Q: How did you validate referential integrity in your model?

A: I validated it in two stages. First, in Power Query before building relationships, I used a Merge Queries left outer join from each fact table to its dimension on the key column, then filtered for nulls in the dimension's return column — zero null rows means every foreign key has a matching primary key. I repeated this check for all 19 relationships. Second, after building the relationships in Model View, I built a simple test table visual with year and total production, confirmed it returned six rows with plausible non-zero values, and spot-checked several specific cells against manually calculated figures from the raw CSV. I then documented both validation stages in Volume 2 of the project documentation as a formal sign-off checklist.

6.2 DAX Questions

Q: Explain the difference between a calculated column and a measure.

A: A calculated column is computed row-by-row at data load time and stored as a physical column in the model — it doesn't change based on how a report is filtered. A measure is computed dynamically at query time in response to the current filter context, which means it can respond to slicers, visual filters, and cross-highlighting. The general rule I follow is: if a calculation needs to respond to filters, it's a measure. If it's a fixed, row-level attribute that never changes with filtering, it can be a Power Query custom column — which is actually even better than a calculated column because it's computed before the data enters VertiPaq, keeping the model leaner. In this project, I used that logic to add Shutdown Duration Category, Employee Tenure Band, and Incident Severity Weight as Power Query columns rather than DAX calculated columns.

Q: Why do you use DIVIDE() instead of the division operator in all your measures?

A: The division operator in DAX returns a hard error when the denominator is zero — which in a filtered report happens frequently, since slicing to a period with no data is common. DIVIDE() handles division by zero gracefully by returning BLANK() instead of an error, which means KPI cards and chart cells simply appear empty rather than showing a red error. In a dashboard seen by executives or external stakeholders, a formula error visible on screen is a significant credibility problem. DIVIDE() is essentially a safety measure that I apply to every ratio calculation in the model as a non-negotiable standard.

Q: How do your time intelligence measures work, and what does the model need for them to function correctly?

A: Power BI's native time intelligence functions — TOTALYTD, SAMEPERIODLASTYEAR, DATEADD, and similar — require a continuous, gap-free date column that is explicitly marked as a Date Table. In this project, dim_calendar was engineered to cover every calendar day from 1 January 2019 to 31 December 2024 with no missing days, and I marked it as a Date Table in Volume 2's modeling step. That marking tells Power BI's engine to enable time intelligence against that table. My YoY measures use SAMEPERIODLASTYEAR on dim_calendar[Date], my YTD measures use TOTALYTD, and the first year in the dataset — 2019 — correctly returns BLANK for all YoY measures since there is no prior year to compare against.

Q: Tell me about the most complex DAX measure you wrote in this project.

A: The most technically interesting measure is Estimated Revenue Lost to Shutdowns. It combines three fact tables conceptually: it takes the lost production volume from fact_shutdown_events, and multiplies it by an average price per tonne calculated from fact_export_shipments. The important DAX detail is that I used ALL(dim_calendar) inside the price calculation to strip out the current date filter when computing the average price — if I used the filtered price, it would create a self-referential estimate that changes unpredictably as the user changes the date filter. By using the all-time average price instead, the measure gives a stable, comparable estimate regardless of which period is selected. This was a deliberate design choice I documented specifically in Volume 3, and it is the kind of nuance that demonstrates genuine understanding of filter context rather than just formula syntax.

6.3 Dashboard Design Questions

Q: How did you decide what visuals to use on each dashboard page?

A: I built a visual selection framework in Volume 4 that maps data type to visual type — for example, trends over time always use line charts, part-to-whole always uses donut charts with a maximum of five segments, and rankings always use horizontal bars sorted descending. The reasoning is that visual type should be chosen to answer a specific question as efficiently as possible, not for aesthetic variety. I also avoided several common poor choices: no gauge charts (they waste space and are imprecise), no 3D charts (they distort proportions), and no pie charts with more than five segments. On every page I identified the one visual that best tells the primary story — the heatmap matrix on the Production page, the waterfall on the Finance page, the Gantt timeline on the Shutdown page — and designed the rest of the page to support that signature visual.

Q: Why did you put navigation buttons on every page rather than using the standard Power BI pages tab?

A: Two reasons. First, in a published Power BI app or embedded report, the pages tab is often hidden, so navigation buttons are the only way a stakeholder can move between pages. Building the navigation into the report from the start means it works correctly in every delivery context, not just in Desktop view. Second, the navigation strip with a highlighted active page button is a much more professional UI pattern than the native tab — it tells the user where they are and what else is available in a single glance. The small extra build time in Volume 4 pays back immediately every time someone reviews the portfolio file and can navigate it intuitively without instructions.

Q: You mentioned reversing conditional formatting on the HSE dashboard. Can you explain that decision?

A: Standard Power BI conditional formatting defaults to green for high values and red for low values — which is correct for Revenue, Production, and Profit, where up is always better. But for HSE incidents, fewer incidents is better, so a rising incident count should show as red and a falling count should show as green. If I applied the default formatting to the HSE YoY% measure, a 20% increase in incidents would appear green, which would be both factually wrong and potentially embarrassing in a real presentation. I documented this specifically in Volume 4 as the most common HSE dashboard design mistake, and applied manual conditional formatting rules that reverse the convention for every safety-related measure on the HSE page.

6.4 Business Understanding Questions

Q: You used an LNG company as the basis for this project. How much do you actually know about LNG operations?

A: Enough to design a model that a real LNG operator could recognize and use. The dataset reflects genuine industry structure: six liquefaction trains each with independent availability, a product mix of LNG, LPG, condensate, and NGL consistent with how a real liquefaction plant separates its output, planned turnarounds of 21 to 35 days that match the typical real-world maintenance window, and an export customer base spanning long-term Sale and Purchase Agreements with European and Asian utilities alongside spot cargoes sold to traders. The KPI framework — plant availability, utilization, corrective maintenance share, LTIFR — are the metrics that real plant managers in this industry actually track. I built this literacy deliberately by researching NLNG's published reports and industry documentation before designing the dataset, because a domain-literate dataset produces far more realistic and defensible insights than a generically structured one.

Q: What business problem does this dashboard actually solve?

A: Six specific ones, all defined in Volume 1 of the project documentation. The most important is the lack of a unified view linking operational performance to financial outcomes — in a real LNG operation, the production manager and the CFO often cannot see in one place how a Train 3 emergency shutdown affected that quarter's profit. This dashboard solves that by connecting fact_production, fact_shutdown_events, and fact_finance through a shared calendar dimension, so the Estimated Revenue Lost to Shutdowns measure gives an immediate financial consequence figure for any operational event selected on the Shutdown Dashboard. That cross-functional connection between operational and financial data is the core value this project demonstrates.

Q: What would you improve about this project if you had more time?

A: Three things. First, I would add a What-If parameter to the Finance Dashboard to let the CFO run price sensitivity scenarios — what happens to profit if LNG prices drop 10%? That would turn the Finance page from a historical reporting tool into a forward-looking planning tool. Second, I would replace the estimated LTIFR constant with actual man-hours data from an HR system, since the current measure uses a placeholder that I explicitly flagged as a limitation in Volume 3. Third, I would add Power BI's anomaly detection feature to the Production trend line, which uses machine learning to automatically flag unusual production drops — that would make the Production Dashboard a more proactive tool rather than just a retrospective one. I documented all three of these as future enhancements in the project because acknowledging the limitations of your own work honestly is as important as demonstrating what you built.

7. Full Presentation Script

The script below is written for a 15-minute live demonstration of the Project APEX dashboard, suitable for a portfolio review, a job interview demonstration request, or a stakeholder presentation. Adapt naturally to your audience — this is a guide, not a recitation.

7.1 Opening (1–2 minutes)

“Good [morning/afternoon]. I want to walk you through Project APEX, an enterprise analytics platform I built to simulate the full operations of Atlantic LNG Operations Limited, a fictional LNG producer based on Bonny Island, Nigeria. Before I open the dashboard, let me give you 30 seconds of context. An LNG company converts natural gas into liquid form, ships it in specialized tankers to customers in Europe and Asia, and earns revenue per unit of gas delivered. The business runs on two things: plant reliability — are the liquefaction trains running — and commercial performance — are we selling at competitive prices to creditworthy customers. Everything I built in Project APEX is designed to give the people running this company visibility into exactly those two things. Now let me show you.”

7.2 Executive Dashboard (2–3 minutes)

Open Page 1 — Executive Dashboard. Narrate the KPI cards from left to right:

- Plant availability: ‘We can see average plant availability across all six trains for the selected period is [X%]. The target is 90%. Green means we are above it, gold means we are approaching it, red means we are below.’
- Total LNG Produced YTD: ‘This is cumulative output for the year selected. Year to date we have produced [X] metric tonnes of LNG across all trains.’
- Profit Margin %: ‘This is the headline profitability figure from our finance fact table — revenue minus operating cost, capital expenditure, and maintenance cost.’
- Then: ‘Let me change the year slicer to 2020 and 2021 and you can see the trend arrow and conditional colour respond immediately as the filter context changes. That responsiveness is the SAMEPERIODLASTYEAR measure in DAX updating the comparison in real time.’

7.3 Production Dashboard (2–3 minutes)

Navigate to Page 2. Focus on two visuals:

- The heatmap matrix: ‘This is the signature visual on this page. Each cell is one train in one month. The colour goes from green for high output to red for low. You can immediately see which trains had difficult months without clicking on anything — this is what I mean by designing for insight speed.’

- The availability vs utilization area chart: ‘The gap between these two shaded areas is meaningful — it shows periods where the train was available but not fully utilized, which is a different problem from a train that is simply down.’
- Click one train on the ranking bar chart: ‘Right-clicking here gives me a drill-through option to a train-specific detail page showing all shutdowns, maintenance events, and HSE incidents for that train alone.’

7.4 Shutdown & Finance (3–4 minutes)

Navigate to Page 6 — Shutdown Dashboard:

- ‘This Gantt timeline shows every shutdown event across all six trains. Bars coloured blue are planned turnarounds. Red bars are emergency shutdowns. When you see multiple red bars clustered in the same period, that is a reliability signal worth investigating.’
- Point to the Estimated Revenue Lost card: ‘This is the financial consequence measure. It combines lost production tonnes from this fact table with average export pricing from the shipments fact table — two separate fact tables connected through the shared calendar dimension. This is the specific business problem from the original brief: linking operational events directly to financial outcomes.’

Then navigate to Page 4 — Finance Dashboard:

- ‘The waterfall chart shows the journey from total revenue to net profit. You can see exactly where the money goes — how much to OPEX, how much to CAPEX, how much to maintenance — in a single visual. No chart literacy required.’

7.5 Closing (1–2 minutes)

“What I want you to take from this is not that I can build a dashboard that looks good — that is the minimum. What I want to demonstrate is that every design decision in this project was made for a specific business reason, every DAX measure was validated against the underlying data, and the entire project is documented to a standard where another analyst could pick it up from Volume 1 and continue it without needing to ask me a single question. That is what I mean when I say this project represents how I work — not just the output, but the process. Happy to go deeper on any specific page, any DAX measure, or any data modeling decision.”

8. Project APEX Completion Checklist

Use this checklist to confirm that every deliverable across all five volumes is complete before publishing or presenting Project APEX as a finished portfolio item.

8.1 Volume 1 — Project Planning

- Business context and company profile documented
- Business problems clearly defined and mapped to dashboards
- Stakeholder map and RACI matrix complete
- Dataset overview and data dictionary documented
- Star schema design documented and ERD confirmed

8.2 Volume 2 — Data Modeling

- All 14 CSV files imported into Power BI Desktop
- Power Query cleaning checklist completed for all 14 tables
- dim_calendar marked as Date Table
- All 19 relationships built and validated in Model View
- fact_shutdown_events dual-date relationship confirmed (StartDateKey active, EndDateKey inactive)
- Three Power Query custom columns added (ShutdownDurationCategory, TenureBand, SeverityWeight)
- Auto date/time disabled in Power BI options
- Data validation sign-off completed

8.3 Volume 3 — DAX Development

- Seven measure tables created (_Production, _Export, _Maintenance, _Shutdown, _HSE, _Finance, _Weather)
- All 45+ measures written and named per Volume 3 conventions
- USERRELATIONSHIP() measures tested on fact_shutdown_events
- YoY measures confirmed to return BLANK for 2019 (no prior year)
- All DIVIDE() functions confirmed to return BLANK on zero denominator
- Revenue and Profit calculated measures validated row-level
- DAX validation checklist completed

8.4 Volume 4 — Dashboard Design

- Custom Power BI theme JSON created and imported
- All 9 dashboard pages built and formatted per wireframes
- Navigation strip with active-page highlighting on all 9 pages
- Three drill-through sub-pages configured (Train, Customer, Incident)
- Five custom tooltip pages configured
- Year slicer synced across all pages
- HSE Dashboard YoY conditional formatting reversed
- Map Dashboard using Latitude/Longitude columns from dim_countries
- All visuals use Segoe UI font and ALOL colour palette

8.5 Volume 5 — Portfolio & Documentation

- LinkedIn project description written and ready to post
- LinkedIn announcement post written
- GitHub README complete and tested
- All 5 volume .docx files in /docs/ folder
- Interview preparation guide reviewed and answers practiced aloud
- Presentation script rehearsed at least once end-to-end
- Project APEX .pbix file saved and tested on a clean machine

PROJECT APEX — SERIES COMPLETE

Volume 1 through Volume 5

"The mark of a complete portfolio project is not that it exists — it is that you can explain every decision in it, defend every number in it, and tell the business story it was built to tell. Project APEX does all three." — Allwell Hart

End of Volume 5 — Portfolio, Documentation & Interview Preparation